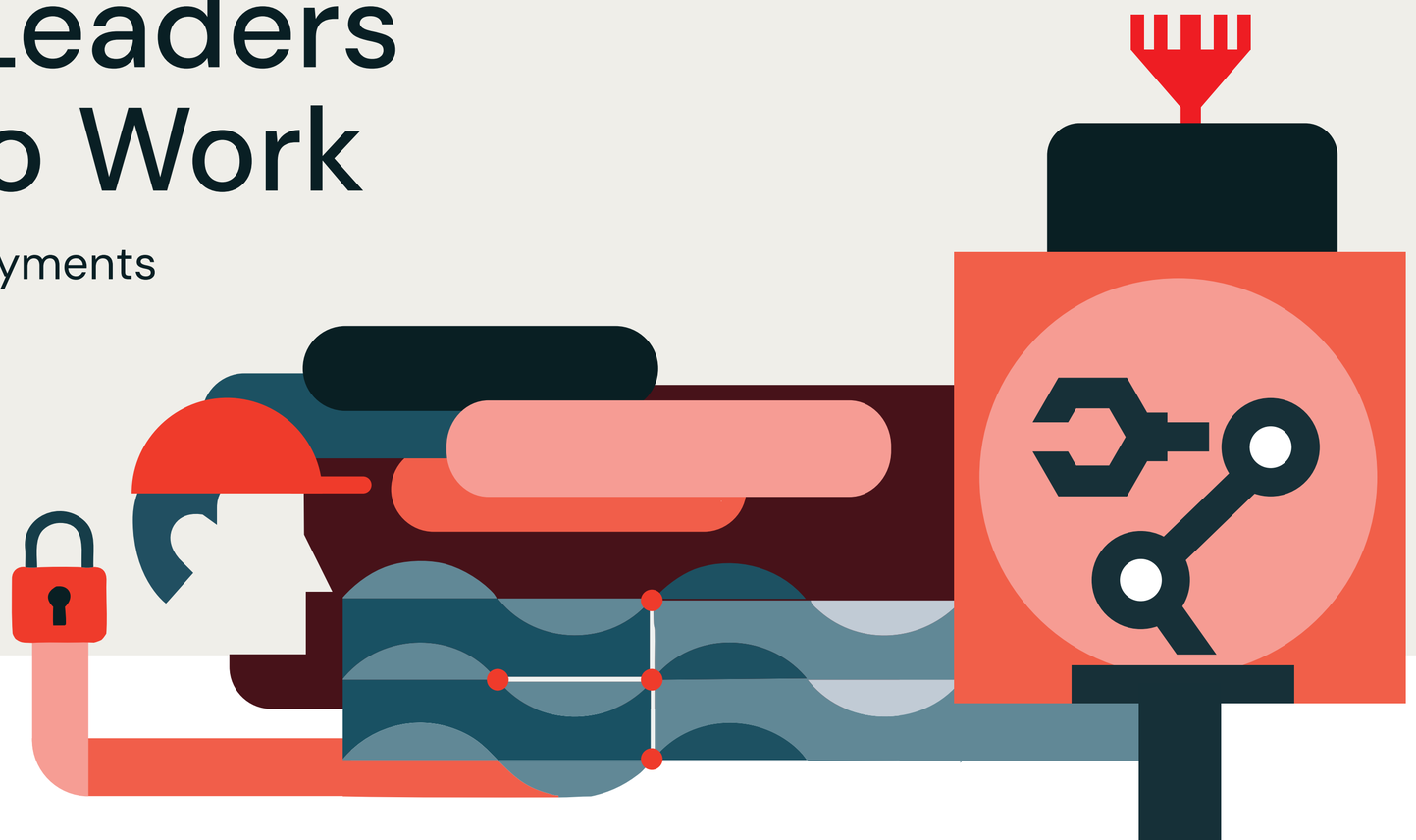
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EXECUTIVE BRIEF

How Industrial Leaders Are Putting AI to Work

Based on Hundreds of Databricks Deployments





Introduction

Industrial manufacturers, logistics providers and distributors sit on enormous volumes of operational data — spread across dozens of systems, locked in PDFs, trapped in equipment sensors and siloed by department. As a result, decisions that should take minutes take days. Analyst teams are buried in repetitive queries. Equipment failures that could have been predicted weeks in advance go undetected until they halt a production line.

The companies pulling ahead aren't simply deploying AI; they're deploying AI on a unified, governed data platform. That's the difference between a pilot that impresses and a system that scales.

Key takeaways

Multi-agent architectures, not single-purpose chatbots, are the production standard for leading companies. BASF Coatings' Marketmind uses a three-tier agent hierarchy — supervisor, manager and specialized workers — to serve 1,000+ sales reps from a single governed platform.

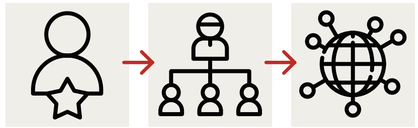
Data governance is the architectural prerequisite for industrial AI at scale. Unity Catalog's fine-grained access controls, column-level security and full lineage through agent workflows are what allow enterprises like BASF to deploy AI globally without losing control of sensitive data.

The most effective industrial AI works across every data type — structured and unstructured — in a single coordinated system. For example, SQL and Genie agents handle database queries while RAG and Vector Search agents tackle PDFs and technical specs. As a result, a sales rep can ask a question and get an answer that pulls from both a pricing database and a 200-page product doc, without having to know where the data lives.

Time to value is measured in weeks, not quarters. Across Databricks deployments, companies are reaching production faster than expected: easyJet delivered a working MVP in two weeks; Edmunds deployed RAG applications in weeks; BASF Coatings moved from pilot to enterprise-scale production in six months.

The efficiency gains are real and attributable. Gerdau cut data processing time from 1.5 hours to 12 minutes and reduced costs by 40%. Edmunds reduced per-model analytics effort by 30%.

Align your stakeholders before you build. The most successful deployments involve CDOs, CTOs, VP sales, VP Ops, IT security and finance from day one. This cross-functional collaboration is what separates pilots that scale from pilots that stall.









87% Faster Processing

40% Reduced Costs

30% Reduced Effort

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Use cases

Multi-agent supervisor architectures

Most industrial companies rely on dozens of systems spread across databases and document types. Multi-agent architectures solve this by assigning specialized agents to handle different tasks, all managed by a central supervisor that routes each question to the right place. BASF Coatings uses this approach to power Marketmind, a sales intelligence platform deployed to over 1,000 reps globally through Microsoft Teams. This solution provides a single governed entry point for answers that used to require multiple systems and manual searches.

Built on: *Unity Catalog (access controls and data lineage), Genie agents (structured data/SQL), RAG agents (unstructured documents), function-calling agents (vector search), Databricks Model Serving*

Natural language data access and analytics

What if anyone on your team could get answers from your company's data just by asking a question — no SQL, no analyst queue, no waiting? easyJet built a text-to-SQL chatbot in two weeks using Whisper for speech transcription and open-source LLMs on Databricks, cutting time to insight from hours to minutes and enabling business users to query complex datasets through conversation alone.

Edmunds went further with DataDave, an analytics agent that runs every query through a five-phase validated workflow: Triage → Planning → Code generation → Execution → Synthesis. Each phase is checked by a specialized critique agent before proceeding, resulting in 95% accuracy on complex analytics queries and proactive insights.

Built on: *Databricks Model Serving, open-source LLM support (Llama, Mistral), Genie text-to-SQL, Mosaic AI critique agent framework*

Predictive maintenance and digital twins

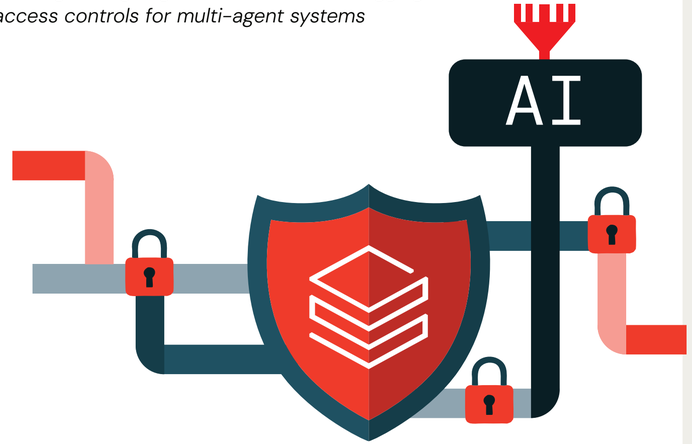
Unplanned downtime is one of the most expensive problems in manufacturing, but predictive maintenance uses real-time data from equipment sensors and production lines to flag issues before they become failures. To put this into practice, Gerdau built digital twins of their steel production lines, cutting data processing time from 1.5 hours to just 12 minutes while reducing data processing costs by 40%.

Built on: *Databricks streaming pipelines, Delta Lake, real-time ML model serving, Mosaic AI*

Data governance as enabler

AGovernance is what makes enterprise-wide AI deployment possible by giving organizations the confidence to scale. Unity Catalog's access controls, data masking and lineage tracking are built into the platform, so AI agents can operate across sensitive datasets without creating compliance risk. BASF Coatings maintains strict data protection standards while empowering AI-driven insights across its global sales team.

Built on: *Unity Catalog (column-level security, dynamic data masking, data lineage tracking through agent workflows, audit logging), fine-grained access controls for multi-agent systems*



Results from the field

Three patterns emerge across these deployments

First, the **efficiency gains** are concentrated in data operations: **processing, querying and enrichment**. This repetitive, high-volume work is exactly what AI handles best.

Second, the **speed improvements** are transformational, not incremental. **Gerdau's 87% reduction in processing time** and **easyJet's hours-to-minutes insight cycle** fundamentally change how fast teams can act on information.

Third, **scale compounds** — each AI workflow absorbs more throughput without adding operational cost. **Grainger** isn't handling **400,000 daily product updates** with proportionally more people; the platform handles volume that would be impossible to manage manually.

In every case, the underlying enabler is the same: Databricks Unified Data Platform that connects raw operational data to the AI layer without requiring teams to stitch together point solutions to bridge the gap.

GERDAU	87%	Faster Processing 1.5 hours → 12 minutes
easyJet	Mins	Faster Time to Insight Hours → Minutes
edmunds	95%	Query Accuracy Complex analytics queries
GRAINGER	400k	Daily Updates AI-managed product catalog

Strategic recommendations

Build your foundation

Before your first agent deployment, implement Unity Catalog. Its built-in access controls, data lineage and audit logging are what allow you to expand AI beyond your first team without creating governance debt. Start by auditing your manual, repetitive data workflows (report generation, data quality checks, catalog updates) and flag the highest-volume ones as your first automation targets. Then, run a focused two-week pilot on a real problem, as easyJet did with their text-to-SQL MVP.

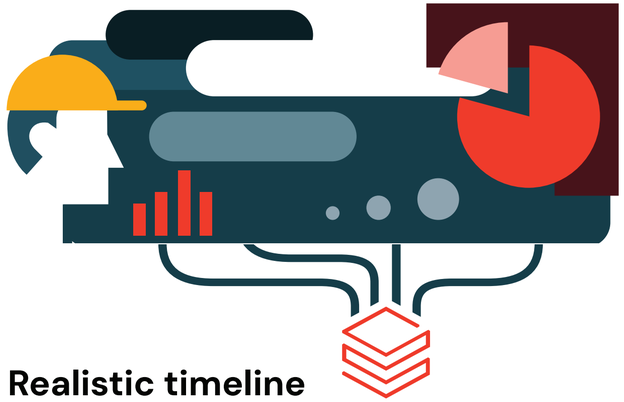
Scale what works

Once a workflow is proven in one business unit, treat it as a template for the next. Creating each new deployment from scratch slows rollout and introduces unnecessary risk. Databricks enables reusable pipelines, shared notebooks and governed data assets that let you take a text-to-SQL workflow or multi-agent architecture and expand it across an organization without rearchitecting. Use MLflow to track model performance and Unity Catalog to manage access as new teams and data domains come online. The goal is to build a library of reusable agent components instead of a collection of isolated deployments.

Skip what doesn't

Not every investment is worth making. Here are three to avoid:

- DIY MLOps infrastructure. Databricks and major cloud providers have already solved this; building your own means spending engineering time on maintaining infrastructure instead of solving business problems. Buying the platform allows you to focus on deploying AI where it impacts operations.
- Single-purpose AI point tools. Every additional vendor increases the total cost of ownership and adds integration debt.
- Premature model optimization. LLMs improve every quarter. Build on an open architecture that lets you swap in better models as they become available without rebuilding your workflows.



Realistic timeline

When it comes to AI, business decision-makers often wonder: How long before this technology delivers something real?

Based on Databricks deployments across industrial companies:

~2 weeks to a working MVP at easyJet, where business users went from zero to querying live data through a text-to-SQL chatbot

2–6 weeks on average for production-ready prototypes at Edmunds, including RAG applications and a validated analytics agent with 95% query accuracy

~6 months for the enterprise-scale rollout of Marketmind at BASF Coatings, from the initial pilot to a governed platform serving 1,000+ sales reps globally

Real, measurable value can be delivered in weeks, not the multi-year transformation cycles most enterprises are used to. The companies that move fastest have two things in place before they start: a governed data foundation, so agents have clean, accessible data to work with, and a clear business owner who defines what "working" looks like. Both are achievable before your first sprint begins.

Conclusion

Industrial AI is no longer an experiment; it's an operational capability that leading manufacturers, distributors and logistics providers are deploying at scale right now. The companies pulling ahead aren't doing it with a collection of point solutions or a single chatbot. They're doing it with a unified platform that brings data engineering, governed access, multi-agent orchestration and model serving together in one environment.

Databricks is the defensible choice for industrial AI at scale, and it comes down to three advantages:

First, open architecture: Databricks provides native support for ChatGPT, Claude, Llama and other open-source models, so there's no model lock-in, and you never have to rebuild when a better model emerges.

Second, governance at the platform layer: Unity Catalog makes it possible to deploy AI broadly across an organization without compromising data protection or compliance.

Third, the compounding advantage of a unified platform: Every new use case built on Databricks shares the same data assets, pipelines and governance framework, so the tenth deployment takes a fraction of the effort the first one did.

The question for industrial leaders isn't whether to deploy AI. It's whether to spend the next two years stitching together tools that don't scale, or build on a unified foundation where every deployment makes the next one faster, smarter and more efficient.

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